Expense Tracker

AI-built personal finance tracker and product vision for an AI-assisted analytics SaaS

What Claude Code Produced Well

Clear learning narrative The notes explain not just what was built, but why it matters: personal pain, then a path to a useful product.	**Portfolio-grade stack** Supabase, FastAPI, dbt, and Power BI are practical tools that map well to data engineering interviews and demos.	**Strong AI framing** The split between AI as builder now and AI as product feature later makes the project easy to explain.
Good production context The notes connect this practice app to Fabric, Synapse, Databricks agents, tool calling, and AEST handling.	**Useful roadmap** Done and pending sections make it obvious what is complete and what should become the next sprint.	**Concrete debugging history** The issue log captures real implementation learning: API keys, DNS, RLS, CLI install, and Git author fixes.

Recommended Improvements

- Fix date freshness: PROJECT_NOTES says last updated 2026-05-07, but the PDF was created on 2026-06-10.
- Clarify repository naming: notes mention finflow-ai and the DS monorepo; make the public demo repo story explicit.
- Avoid saying zero manual coding required unless presenting it as a learning reflection; it can sound less credible than AI-paired implementation.
- Move secrets detail out of showcase material. It is fine to say legacy anon key compatibility was handled, but do not dwell on key format in a public PDF.
- Use one polished generated image instead of many tiny hand-drawn PDF shapes; it communicates the product faster.

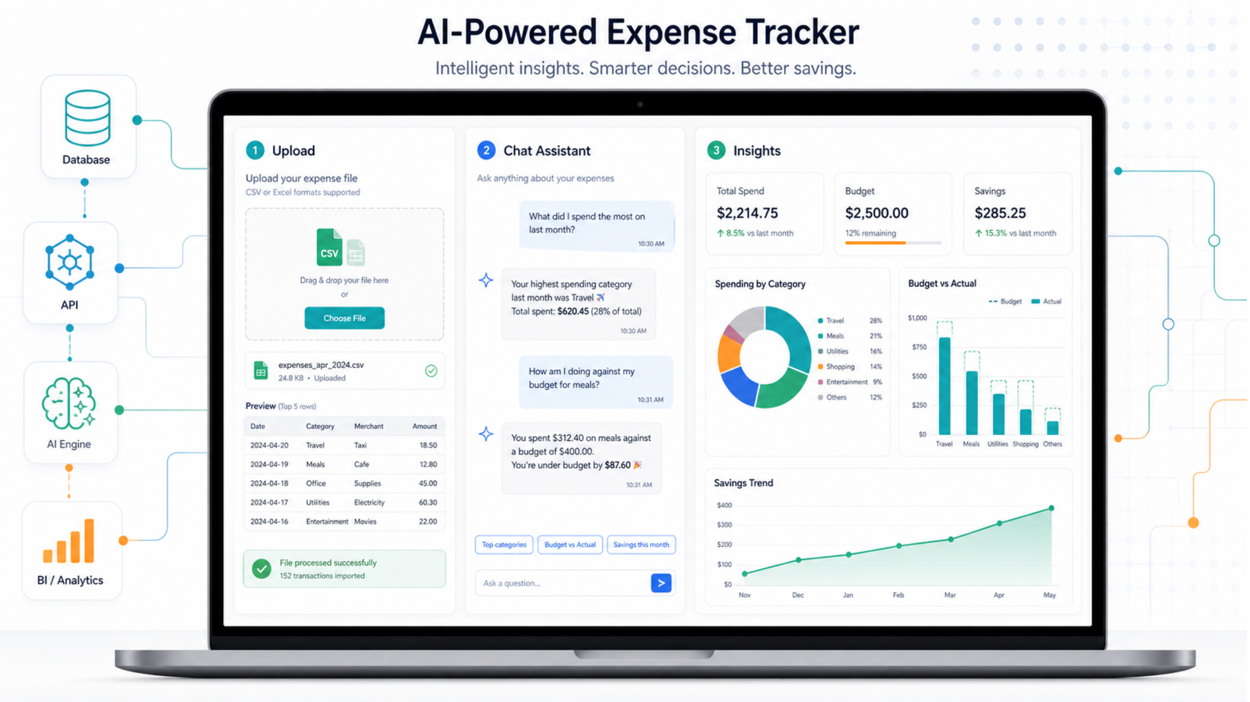
Core Takeaways

1. AI as co-builder Claude helped design, code, debug, and explain the stack.	**2. Real data stack** The project uses tools that transfer to enterprise data work.
3. Personal pain to product A household tracker becomes a SaaS idea for small businesses.	**4. Agent-ready architecture** FastAPI functions can become tools for natural-language analytics.

Best presentation angle: 'I used AI to build a real data product, then designed the next step where AI becomes the product interface.'

AI-Powered Product Vision

A cleaner showcase visual generated with GPT Image-2 and embedded into this revised PDF



How to Explain the Architecture

1. Upload Customer drops a CSV or Excel file. AI detects columns, normalises data, maps categories, and flags messy rows.	**2. Transform** FastAPI writes clean data to Supabase. dbt models produce monthly summaries, savings rate, and budget variance.	**3. Ask** The chat agent calls Python tools such as get_monthly_summary, compare_months, and get_budget_status.
4. Visualise Power BI embedded shows the dashboard and can be filtered from natural-language questions.	**5. Scale** The same pattern can support restaurant or retail owners who have spreadsheets but no data team.	**6. Differentiate** This is not only a budget app; it is an AI data-cleaning and analytics workflow for non-technical users.

Customer Data Flow

CSV / Excel	AI Cleaner	Supabase	dbt	Power BI + Chat
Raw uploads	Column mapping and categories	Clean relational data	Analytics marts	Insights for users